

Data of Ship Block Scheduling in English

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Description

This study is based on the following assumptions:
Assumption 1: The projection shape of each block position and block is a minimum bounding rectangle.
Assumption 2: Each block position can process only one block at a time.
Assumption 3: Labor force is sufficient, without absenteeism or other anomalies.
Assumption 4: Blocks of different sizes can find suitable positions for processing.
Assumption 5: Each block is completed at only one block position.
Assumption 6: All block positions are empty at the beginning of scheduling.

The dataset includes input data of blocks and block positions, scheduling results of EDDQN, DDQN, and five heuristic algorithms, as well as raw data for figures in the paper.

A key observation is that the EDDQN algorithm shows stronger advantages in large-scale scheduling problems than in small- and medium-scale cases.

The data reflects scheduling performance under the stated assumptions. Lower makespan, lower cost, and fewer position switches indicate better scheduling performance. Users can load the input data, run the provided algorithms, and reproduce the experimental results and figures in the paper.

```
Folder Structure
data_en/
├── Raw_input_Data/
│   ├── block_information_table.xlsx
│   ├── initial_block_position_information.xlsx
│   └── block_position_information.xlsx
├── Result/
│   ├── DDQN_scheduling.xlsx
│   ├── EDDQN_scheduling_results.xlsx
│   ├── scheduling_results_heuristic_earliest_start_time_20250605_014601.xlsx
│   ├── scheduling_results_heuristic_longest_processing_time_20250605_014601.xlsx
│   ├── scheduling_results_heuristic_resource_utilization_based_20250605_014601.xlsx
│   ├── scheduling_results_heuristic_response_time_based_20250605_014601.xlsx
│   └── scheduling_results_heuristic_shortest_processing_time_20250605_014601.xlsx
├── Figure6&7_raw_data/
│   ├── DDQN_results.xlsx
│   ├── EDDQN_results.xlsx
│   └── Figure10_raw_data.xlsx
└── mapping_key_tests.csv
```

1) Input Data ("Raw_input_Data")

block_information_table.xlsx
Block-level production and scheduling attributes.

Typical fields include:
- block identifier / sequence,
- vessel identifier and vessel type,
- block type,
- dimensions and weight,
- processing time,
- earliest/latest start time and latest completion time,
- combination-related attributes (if applicable).

UserMain job-side input for scheduling models.

"initial_block_position_information.xlsx"
Initial assignment status between blocks and block positions.
Typical fields include:
- vessel/block identifiers,
- earliest start time,
- initial block-position description,
- expected completion time.

Use: Defines initial state and initialization constraints.

block_position_information.xlsx
Resource and spatial constraints for block positions.
Typical fields include:
- block-position ID and description,
- area hierarchy,
- size/capacity limits,
- lifting capacity and height constraints,
- labor-team attributes,
- designated/general usage attributes.

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Files

 Figure6&7_raw_data	
 Raw_input_Data	
 Result	
 Figure10_raw_data.xlsx	13.9 KB ↓
 mapping_key_tests.csv	3.52 KB ↓
 README.md	6.72 KB ↓

Steps to reproduce

The data in this dataset is adapted from real-world shipyard production data, with minor adjustments to ensure anonymity and experimental controllability.

Institutions

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Shanghai, Shanghai

Categories

Machine Learning, Large-Scale Optimization, Ship Production Technology

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